[AA] Project Assignment 2025/26

Analytics and Applications, M.Sc. Course
Faculty of Management, Economics, and Social Sciences
Department of Information Systems for Sustainable Society
University of Cologne

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This project assignment is designed to test a representative cross-section of the data analytics and machine learning approaches we cover during this course. It is based on a real-world problem with high relevance to the current hot topic of electric vehicles (EVs) and will act as an illustration of how we can use data in impactful ways to address societal issues and business at the same time.

Please download all relevant data from Sciebo: `https://uni-koeln.sciebo.de/s/C5ewJWCRNCoboPS`

1 Introduction

Transport-related greenhouse gas emissions make up for the second-largest chunk of total EU emissions. It has thus long been recognized that in order to meet decarbonization targets, our approach to mobility will have to change. To this day, traditional urban mobility relies primarily on internal combustion engine (ICE) vehicles. A switch to EVs is considered a key ingredient in reducing personal mobility emissions. However, the charging infrastructure and underlying power grid need to provide appropriate capacity. With EVs, many users are experiencing what is widely called "range anxiety" - the fear of not being able to complete a trip due to uncertain battery capacity and the fact that EVs cannot be recharged instantaneously, like ICE cars can. While this is largely psychologically driven and has a lot to do with experience and will likely fade over time as we get to know EVs better (and EV technology will allow for longer trips and faster recharge times), it adds to the strain on the power grid, as users would rather plug their EV in when it is parked than not, just to "top it up". What is more, when many users want to charge in the same spot (e.g. in a shopping center), they are capacity constrained, as usually not all charging stations in one hub can operate at full capacity at the same time. Thus, for operators of such charging hubs, it is of great importance to monitor, understand, plan, and potentially even steer the charging sessions of connected EVs. In this project, you will take the role of a team of data scientists that works for the operator of charging hubs. You will carry out all steps of the cross industry standard process for data mining (CRISP-DM) and focus on two core aspects:

1. **Understanding System Metrics:** Gaining a deep understanding of key metrics of the charging hubs' operation and presenting relevant metrics in a refurbished way for the management to understand.
2. **Prediction of Utilization:** Accurately predicting future utilization is an important tool to plan operation and facilitate new business models.

2 Description of Datasets

You have been provided with data on individual EV charging sessions at two charging sites that belong to the same operator. One site is public (at a university), the other is private (open to employees of a company). Both have approximately 50 EV charging stations that are part of the monitored charging network. The datasets have been preprocessed and are provided in CSV format to take some data wrangling off of your shoulders but have not been checked for erroneous or missing data. Additionally, we provide you with hourly historical weather data from a nearby airport weather station. You are free to use this data and additional data sources you might want to include for your analysis. Some charging sessions were carried out by registered users, some by unregistered users. Registered users are able to provide requests to the charging management system, which the system can use to facilitate adaptive charging, but requests are not guaranteed to be fulfilled.

field	type	description
id	string	Unique identifier of the session record
connectionTime	datetime[a]	Time when the EV plugged in.
disconnectTime	datetime[a]	Time when the EV unplugged.
doneChargingTime	datetime[a]	Time of the last non-zero current draw recorded.
kWhDelivered	float	Amount of energy delivered during the session.
sessionID	string	Unique identifier for the session.
siteID	string	Unique identifier for the site.
spaceID	string	Unique identifier of the parking space.
stationID	string	Unique identifier of the charging point.
timezone	string	Timezone of the site. Based on pytz format.
userID	string	Unique identifier of the user, if provided.
userInputs	list	Inputs provided by the user. Since inputs can be changed over time, there can be multiple user input objects in the list.
WhPerMile[b]	float	Efficiency of the EV in Wh per mile.
kWhRequested[b]	float	Energy requested by the user in kWh.
milesRequested[b]	float	Number of miles requested by the user.
minutesAvailable[b]	float	Length of the session as estimated by the user.
modifiedAt[b]	datetime[a]	Time this user input was provided.
paymentRequired[b]	bool	If the user was required to pay for this session.
requestedDeparture[b]	datetime[a]	User estimated departure time.

[a] All datetimes are in UTC (GMT); see timezone field for the correct timezone of the site.
[b] Fields are optional, if user participates in app-based charging.

Table 1. Description of Fields of Charging Session Dataset

3 Task Description

1. **Data Collection and Preparation:** You have been provided with a full dataset of charging sessions in two sites for an extended timeframe. The first step should be to load the data in appropriate format, checking for missing or erroneous data, and cleaning your dataset for use in later stages of your project. Briefly describe how you proceeded and how you dealt with possible missing/erroneous data.
2. **Descriptive Analytics:** The operator of the two sites is interested in the operational performance and statistics of their charging hubs. As the company's data scientist, your task is to facilitate this. Proceed as follows:
   (a) Key Performance Indicators (KPIs): Define three time-dependent KPIs that you would include in a dashboard for the hub operator. These KPIs must provide an immediate overview of the current hub operation and how it is doing in terms of utilization or other business-related aspects. Briefly explain the rationale behind selecting each KPI, explain why you have chosen it and, where needed, provide references. Calculate hourly/daily/monthly etc. values for the selected KPIs and visualize them over time. Which trends and patterns do you observe? How do you explain them?
   (b) Site Characteristics: The hub operator provided you with the dataset, but he forgot which site was supplying which data... Can you find out which of the two sites is the public one? Try to combine data understanding from previous descriptive analytics with domain knowledge (business understanding) of how private vs. public charging hubs might differ in operation. Explain your line of thought!
3. **Cluster Analysis:** To better understand your customer base, carry out a cluster analysis to provide management with a succinct report of archetypical registered users. To do this, create a user-level dataset containing relevant information on your users' charging behavior and implement one clustering algorithm. Try to come up with names for the resulting clusters.
4. **Utilization Prediction:** The operator has asked you to support the data-driven development of a new business case. To do this, proceed as follows:
   (a) **Think of a business case** that would be enabled by a prediction of site utilization. Clearly explain how a prediction model would facilitate this business case. *Simple example: if we know the future utilization of our site, we could run targeted promotions to our registered users in low-demand hours to increase utilization.*
   (b) **Develop a prediction model** that can enable your proposed business case. To do this, choose one of the two sites in the dataset, define a suitable metric for utilization, generate a dataset

at the site-hour level, and use cross-validation to train a prediction model and assess predictive performance. Make sure to motivate your choices. *Simple example continued: we use a neural network to predict the hourly number of charging sessions at Site 1.*

(c) **Give a concrete example** using the data and/or your generated predictions to illustrate how your predictive model enables your proposed business case. Your boss is rather cautious when it comes to money. Sketch up limitations and think about potential pitfalls. In other words: Pitch the viability of your idea, keeping in mind all eventualities! *Simple example continued: we can predict with high certainty that utilization is particularly low on Saturday mornings and on Sundays when it's rainy. If we can increase utilization in these periods by sending targeted promotions, we could boost utilization by X%, assuming that...*

4 Deliverables

Please adhere to Section 2 in the Syllabus, as all deliverables and respective deadlines are detailed there. In addition, we highly recommend you to make use of version-control systems such as git and use platforms like GitHub. The submission process is facilitated via ILIAS:

1. For the **two milestone submissions**, you can either upload your jupyter notebook(s) via the respective upload point in the Project Assignment folder in ILIAS *or* in case you use GitHub, you can invite Julius Kuhmann (kuhmann@wiso.uni-koeln.de) to your repository before the respective milestone deadline. **Remember: Failure to provide milestone progress equals failure of the project assignment component of the combined exam, which, in turn, equals failure of this course!**
2. For the **final submission**, the project report must be submitted via the respective upload point in the Project Assignment folder in ILIAS. The code, again, can either be uploaded via the same ILIAS upload point *or* by sharing your GitHub repository before the deadline.